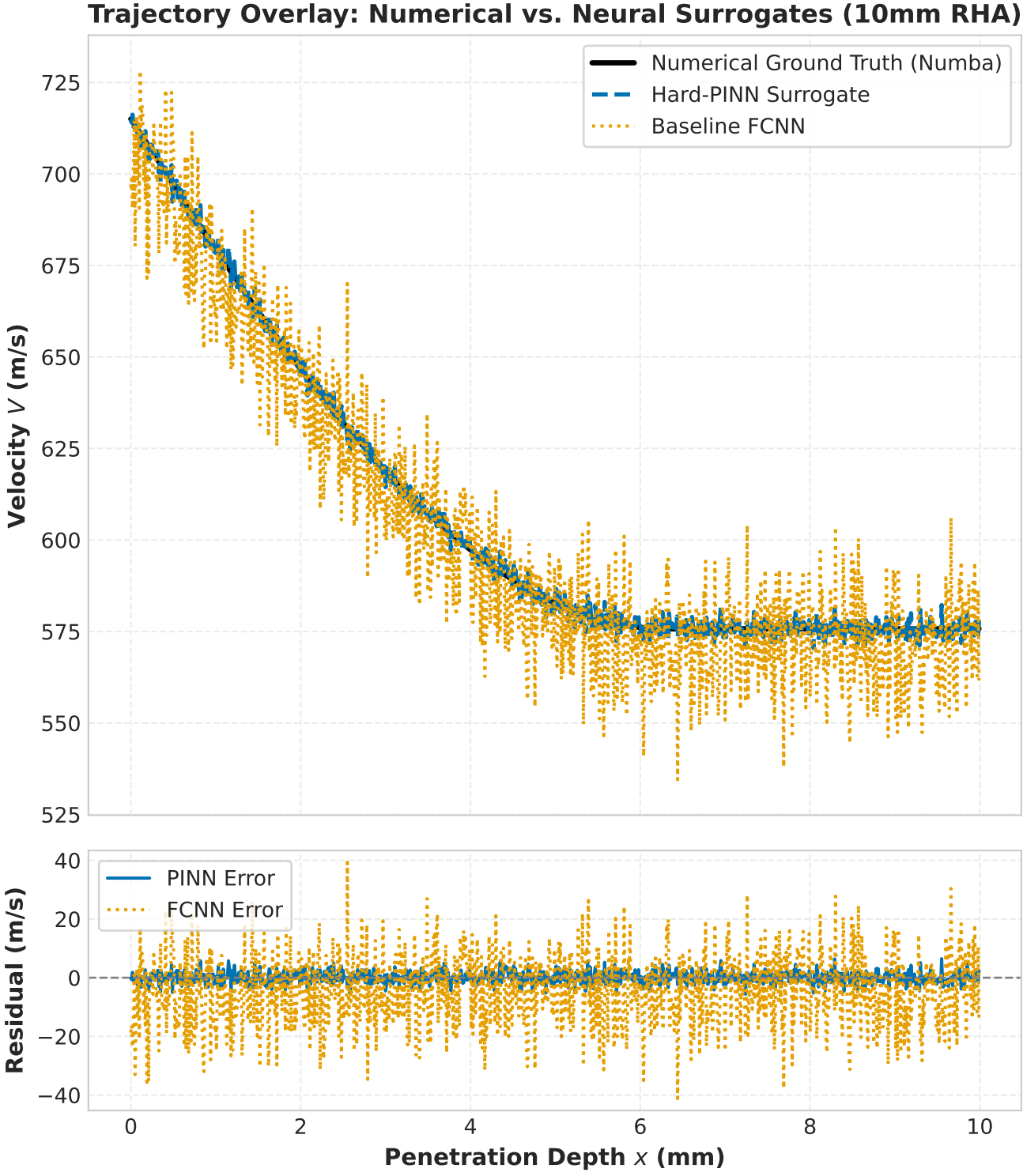

Trajectory Overlay: Numerical vs. Neural Surrogates (10mm RHA)
Numerical Ground Truth (Numba)
Hard-PINN Surrogate
Baseline FCNN
Velocity V (m/s)
725
700
675
650
625
600
575
550
525
PINN Error
FCNN Error
Residual (m/s)
40
20
0
−20
−40
0
2
4
6
8
10
Penetration Depth x (mm)